

EVOLUTION

1. Best fossils are obtained from

- (1) Sedimentary rock
- (2) Igneous rock
- (3) Metamorphic rock
- (4) Any type of rock

2. Which of the following is not a reptilian character of Archaeopteryx ?

- (1) The bones are not pneumatic
- (2) Presence of weak sternum
- (3) An intimate fusion of skull bones
- (4) Jaws bear teeth

3. Read the following statements (A – E)

(A) Presence of a short tail in newborn babies is an example of atavism

(B) The rate of appearance of new forms has no link to the life span

(C) Those animals which are better fit undergo natural selection

(D) Evolution is not a directed process in the sense of determinism

(E) According to Darwin, evolution is a rapid, single-step process

How many of above statements are correct ?

- (1) Two
- (2) Three
- (3) Four
- (4) Five

4. Who was the first hominid ?

- (1) Homo habilis
- (2) Homo sapiens
- (3) Australopithecus
- (4) Dryopithecus

5. At a particular locus frequency of 'A' allele is 0.6 and of 'a' is 0.4. What would be the frequency of heterozygotes in a random mating population at equilibrium ?

- (1) 0.36
- (2) 0.16
- (3) 0.24
- (4) 0.48

6. Pouter, Jacobin and Fantail are the breeds of A produced by B

Choose the correct option for A and B

	A	B
(1)	Fowl	Natural selection
(2)	Leghorn	Hybridisation
(3)	Rock pigeon	Artificial selection
(4)	Fringilla	Adaptive radiation

7. Which of the following is the correct sequence of events in the origin of new species according to Darwin ?

- (1) Natural selection, variation and their inheritance, survival of fittest, struggle for existence, high rate of reproduction
- (2) Enormous fertility, variation and their inheritance, struggle for existence and survival of fittest
- (3) High rate of reproduction, struggle for existence, variation and their inheritance, survival of fittest, natural selection
- (4) Survival of the fittest, variation and their inheritance, struggle for existence, enormous fertility

8. Unit of natural selection is _____ and unit of evolution is _____

- (1) Individual, population
- (2) Population, individual
- (3) Species, population
- (4) Individual, individual

9. Choose the true one with respect to human evolution

- (1) Neanderthal and Homo habilis were living at the same time
- (2) Australopithecus was living in Australia
- (3) Homo erectus had a larger brain capacity than Neanderthal man
- (4) Extinct human ancestor who ate only fruits and hunted with stone weapons was Australopithecus

10. Match the columns and choose the correct option

Column-A		Column-B	
(i)	Homo habilis	(a)	Religious
(ii)	Homo erectus	(b)	First tool maker
(iii)	Homo sapiens neanderthalensis	(c)	Cave painting
(iv)	Homo sapiens fossilis	(d)	First use of fire

- (1) (i)-(b), (ii)-(a), (iii)-(c), (iv)-(d)
- (2) (i)-(d), (ii)-(c), (iii)-(b), (iv)-(a)
- (3) (i)-(b), (ii)-(d), (iii)-(a), (iv)-(c)
- (4) (i)-(c), (ii)-(a), (iii)-(b), (iv)-(d)

11. Which of the following cannot be considered as vestigial organ ?

- (1) Plica semilunaris, coccyx
- (2) Splint bone in horse and wisdom teeth in human
- (3) Additional mammae and dense body hair
- (4) Vermiform appendix, panniculus carnosus

12. Which of the following is a missing link ?

- (1) Euglena
- (2) Neopilina
- (3) Balanoglossus
- (4) Archaeopteryx

13. Read the following statements (A - E):

A - S.L. Miller created electric discharge in a closed flask containing CH_4 , H_2 & NH_3 .

B - Miller observed the formation of Amino acids in flasks like Arginine, Alanine & glutamic acid.

C - The first non-cellular form of life could have originated 3 billion years back.

D - Earth was supposed to have been formed about 4.5 million years back.

E - Primitive atmosphere was reducing.

How many above statements are true ?

- (1) 5
- (2) 4
- (3) 3
- (4) 2

14. Select the correct one w.r.t Jurassic period of the mesozoic era.

- (1) Gymnosperms are dominant plants and first birds appear
- (2) Radiation of reptiles and origin of mammal like reptile
- (3) Dinosaurs become extinct and angiosperms appear
- (4) Flowering plants and first dinosaurs appear

15. Which of the following arrangement of periods of the Paleozoic era gives a correct sequence from earliest to latest ?

- (1) Devonian → Carboniferous → Permian
- (2) Carboniferous → Devonian → Permian
- (3) Permian → Devonian → Carboniferous
- (4) Triassic → Jurassic → Cretaceous

16. The appearance of DDT resistant variety of mosquito amongst the sensitive ones during increased exposure to DDT is an example of

- (1) Disruptive selection
- (2) Stabilizing selection
- (3) Directional selection
- (4) Artificial selection

17. Match the column A with column B & select the correct options:

Column-A		Column-B	
(i)	Wing of Birds & butterfly	(a)	Saltation & result speciation
(ii)	Australian marsupials	(b)	Darwin's laboratory
(iii)	Galapagos island	(c)	Adaptive radiation
(iv)	Single large step mutation	(d)	Convergent Evolution

Options are:

- (1) I - d, II - c, III - a, IV - b
- (2) I - d, II - c, III - b, IV - a
- (3) I - a, II - b, III - d, IV - c
- (4) I - a, II - b, III - c, IV - d

18. Dryopithecus was more _____ and Ramapithecus was more _____ primates who existed about _____ million years ago

- (1) Ape like, man like, 15
- (2) Man like, ape like, 25
- (3) Ape like, man like, 2
- (4) Man like, ape like, 2

19. Find out the incorrect statement

- (1) Hardy-Weinberg equilibrium is applicable to large randomly mating population
- (2) Sewall-Wright effect is applicable to large inbreeding population
- (3) Hybrid inviability causes death of hybrid
- (4) Ethological isolation is due to differences in courtship pattern between closely related species

20. If 64% of the individuals in a population are homozygous recessive for a character then calculate the percentage of heterozygous individuals in this population:

- (1) 36%
- (2) 0.04%
- (3) 0.8%
- (4) 32%

21. Mutations are X while Darwinian variation are Y. X and Y are

- (1) X → Small and directional
Y → Random and directionless
- (2) X → Small and directional
Y → Random and directional
- (3) X → Random and directionless
Y → Small and directional
- (4) X → Random and directional
Y → Small and directional

22. Find the mismatch of different categories of humans with their corresponding character.

- (1) Homo habilis – Brain capacity 650–800 cc
- (2) Australopithecus – Lived in East African grasslands
- (3) Homo sapiens – Arose between 75,000–10,000 mya
- (4) Dryopithecus – Existed about 15 mya and was more like apes

23. Choose the incorrect option

- (1) Evolution of different species of finches in Galapagos are examples of convergent evolution
- (2) Mutations are discontinuous variations
- (3) Small isolated populations are susceptible to genetic drift
- (4) Isolation & natural selection are important factors of evolution

24. Which of the following factor/s do not change the Hardy-Weinberg equilibrium?

- a. Gene migration
- b. Genetic recombination
- c. Random mating
- d. Genetic drift
- e. Gene flow

- (1) a and b
- (2) a, b and c
- (3) c only
- (4) c and e

25. If 64% of individuals in populations are homozygous recessive for a character, then calculate the percentage of individuals in this population who exhibit the dominant trait

- (1) 48%
- (2) 36%
- (3) 4%
- (4) 32%

26. Find the incorrect statement

- (1) The fitness of an individual is referred to as its reproductive fitness
- (2) According to Darwin, any population has built in variation in characteristics
- (3) Darwin highlighted the importance of small and gradual changes in evolution, whereas, de Vries focussed on abrupt and drastic changes as the cause of origin of species
- (4) Evolution at genetic level like sickle cell anaemia is macroevolution

27. Darwin explained that adaptive radiation gave rise to variety of marsupials in Australia in the same process as found in

- (1) Placental mammals in other continents
- (2) Finches in Galapagos islands
- (3) Placental wolf and Tasmanian wolf-marsupial
- (4) Both (1) & (2)

28. Mark the biggest land reptiles which were about 20 feet in height, had huge dagger like teeth and disappeared about 65 mya

- (1) Brachiosaurus
- (2) Stegosaurus
- (3) Tyrannosaurus
- (4) Triceratops

29. Reactants of Urey-Miller experiment include

- (1) Ammonia & methane
- (2) Methane & methionine
- (3) Hydrogen & nucleic acid
- (4) Water & phenylalanine

30. Find the incorrect statement

- (1) Excess use of pesticides resulted in selection of resistant varieties is an example of evolution by anthropogenic action
- (2) Evolution is a stochastic process based on chance events in nature and chance mutation in organisms
- (3) Mutations are random and directional
- (4) Alfred Wallace worked in Malay Archipelago

31. Which of the following is not a feature of mutation theory?

- (1) Mutations may occur in any direction
- (2) Mutations are large and sudden
- (3) Mutations are non-heritable, having no role in evolution
- (4) Mutations are source of discontinuous variations

32. Sudden reappearance of a certain ancestral structure that has completely disappeared or reduced is called

- (1) Atavism
- (2) Convergent evolution
- (3) Divergent evolution
- (4) Speciation

33. At a particular locus, frequency of 'A' allele is 0.3 and that of 'a' is 0.7, what will be the frequency of heterozygotes in random mating population at equilibrium?

- (1) 0.48
- (2) 0.21
- (3) 0.42
- (4) 0.9

34. Which of the following is an example of connecting link species?

- (1) Lobe fish
- (2) Dodo bird
- (3) Sea weed
- (4) Tyrannosaurus rex.

35. Read statements A and B and choose the option with correct statements

Statement A: If the both peripheral characters are selected rather than the mean character it is Disruptive selection

Statement B: This can be characterised by sickle cell anaemia in African population

- (1) A and B
- (2) Only B
- (3) Neither A nor B
- (4) Only A

36. Find the correct sequence of human evolution

- A. Cro-magnon man
- B. Homo erectus
- C. Australopithecus
- D. Homo habilis

- (1) D → A → B → C
- (2) B → C → D → A
- (3) C → B → D → A
- (4) C → D → B → A

37. Fill in the blanks and choose the correct option

a is the unit of natural selection and b is the unit of evolution

- (1) a - population, b - individual
- (2) a - individual, b - population
- (3) a - species, b - phylum
- (4) a - subspecies, b - subphylum

38. Given below are few characters of Archaeopteryx.

Select the reptilian characters

- a. Jaws modified into beak
- b. Fused skull bones
- c. Free caudal vertebrae
- d. Scales present
- e. Non pneumatic bones

- (1) a, b and e
- (2) b, c and d
- (3) b, c, d and e
- (4) c, d and e

39. Read the following statements (A – E)

A. Fossils are one of most acceptable evidences in support of evolution

B. Excreta preserved in form of fossils are called palynofossil

C. Fossils are mostly obtained from sedimentary rocks

D. Elephant like mammoths obtained from Siberia are examples of petrified fossils

E. Most fossils are unpetrified fossils

How many of the above statements are true?

- (1) 1
- (2) 2
- (3) 3
- (4) 4

40. Match Column A with Column B and choose the correct option

Column-A		Column-B	
(A)	Lucy	(i)	Homo sapiens fossilis
(B)	Handy man	(ii)	Australopithecus
(C)	Peking man	(iii)	Homo habilis
(D)	Cro-magnon man	(iv)	Homo erectus

	(A)	(B)	(C)	(D)
(1)	(iv)	(ii)	(iii)	(i)
(2)	(ii)	(iii)	(i)	(iv)
(3)	(ii)	(iii)	(iv)	(i)
(4)	(iii)	(ii)	(i)	(iv)

41. Choose the correct option

Statement A: Mixture of CH₄, NH₃ and H₂ was taken in 2:1:2 ratio in Miller's experiment

Statement B: Alanine, glycine and aspartic acid were obtained at the end of the experiment

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both statements A and B are correct
- (4) Both statements A and B are incorrect

42. Choose the correct option

Statement A Artificial selection is similar to natural selection except that it is an anthropogenic process

Statement B Plants like Cabbage Broccoli Cauliflower etc. has been produced from wild mustard by selective breeding

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both statements A and B are correct
- (4) Both statements A and B are incorrect

43. In a population of 1000 individuals 360 belongs to genotype AA, 480 to Aa and the remaining 160 to aa. Then what will be the frequency of allele A in population ?

- (1) 0.7
- (2) 0.6
- (3) 0.5
- (4) 0.4

44. Mutations are X while Darwinian variations are Y. Here X and Y are

- (1) X → Small and directional
Y → Random and directionless
- (2) X → Small and directional
Y → Random and directional
- (3) X → Random and directionless
Y → Small and directional
- (4) X → Random and directional
Y → Small and directional

45. Adaptive radiation is best shown by

- (a) Darwin finches**
- (b) Host-parasites**
- (c) Australian marsupials**

- (1) a only
- (2) c and b
- (3) c only
- (4) a and c

46. Which of the following statements is correct regarding evolutionary theories?

- (1) Lamarck's theory was supported by Weismann
- (2) Darwinian variations are directionless
- (3) Fitness is the end result of the ability to adapt and get selected by nature
- (4) Theory of Inheritance of Acquired characters was given by Hugo de Vries

47. Fossils provide which type of evidence of evolution?

- (1) Anatomical and morphological
- (2) Embryological
- (3) Palaeontological
- (4) Biogeographical

48. How many of the following statements are false ?

- a** Homo habilis had a brain size of around 900 cc
- b** Homo erectus had a brain size of 650–800 cc
- c** The Neanderthal man had a brain size of 1400 cc
- d** Dryopithecus was more ape like than Ramapithecus
- e** Australopithecus is the connecting link between apes and man

- (1) 2
- (2) 5
- (3) 4
- (4) 3

49. Find the incorrect match

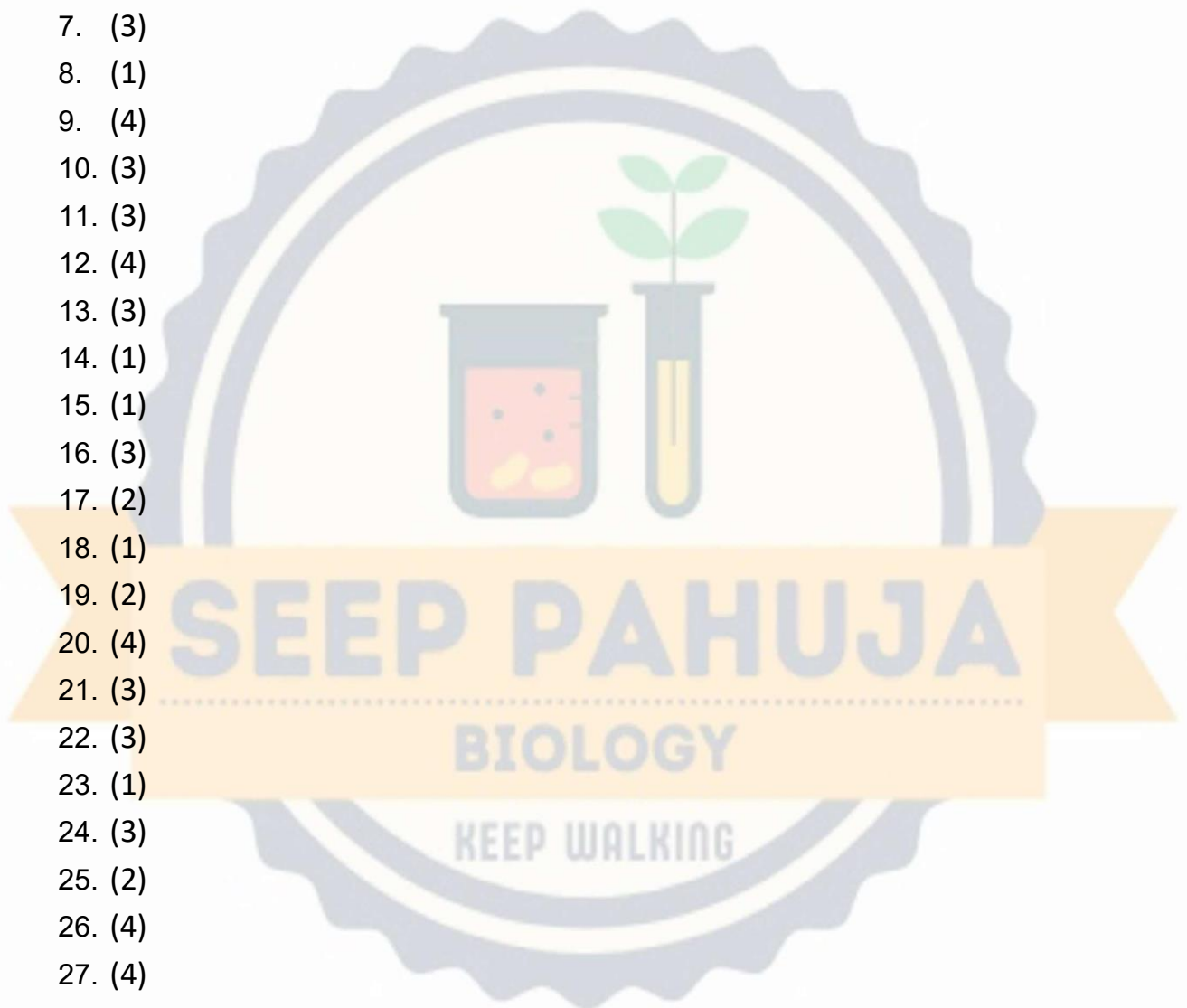
- (1) Lucy – Australopithecus africanus
- (2) Java ape man – Homo erectus erectus
- (3) Skillful man – Homo habilis
- (4) Burial of dead – Homo sapiens neanderthalensis

50. All are examples of living fossils, except

- (1) Neopilina
- (2) Limulus
- (3) Locusta
- (4) Sphenodon

Answer Key

1. (1)
2. (3)
3. (2)
4. (1)
5. (4)
6. (3)
7. (3)
8. (1)
9. (4)
10. (3)
11. (3)
12. (4)
13. (3)
14. (1)
15. (1)
16. (3)
17. (2)
18. (1)
19. (2)
20. (4)
21. (3)
22. (3)
23. (1)
24. (3)
25. (2)
26. (4)
27. (4)
28. (3)
29. (1)
30. (3)
31. (3)
32. (1)
33. (3)
34. (1)
35. (4)



36. (4)

37. (2)

38. (4)

39. (2)

40. (3)

41. (3)

42. (3)

43. (2)

44. (3)

45. (4)

46. (3)

47. (3)

48. (1)

49. (1)

50. (3)

